Title: Design and Implementation of IoT Based Smart Room Automation System for Energy Optimization

Abstract:

This project focuses on designing a smart room automation system using ESP32, PIR sensor, LDR sensor, relay module, and LCD display.

The system automatically controls room lighting based on human presence and ambient light conditions.

Additionally, WiFi connectivity enables manual control through a mobile interface.

The objective is to reduce electricity wastage and introduce students to embedded systems and IoT concepts.

Objectives:

• Automatically switch lights ON/OFF based on occupancy.

• Reduce energy consumption using light intensity detection.

• Enable wireless control via mobile phone.

• Display room status on LCD.

• Provide hands■on learning of microcontrollers and sensors.

Problem Statement:

In many classrooms and rooms, lights remain ON even when no one is present, leading to unnecessary power consumption.

Manual switching is inefficient and unreliable.

This project provides an automated solution using sensors and IoT technology.

System Working:

The PIR sensor detects human motion and sends signal to ESP32.

ESP32 checks ambient light using LDR.

If motion is detected and room is dark, ESP32 activates relay to turn ON the light.

If no motion is detected for a predefined time, lights turn OFF automatically.

LCD displays room status.

User can also control light remotely using WiFi.

Main Components:

• ESP32 Development Board

• PIR Motion Sensor

• LDR Sensor

• Relay Module (5V)

• 16x2 LCD Display

• LED/Bulb

• Jumper Wires and Breadboard

Estimated Budget:

ESP32 – ■400

Sensors + Relay – ■500

LCD + wiring – ■300

Total Approximate Cost – ■1200 to ■1800

Software Used:

• Arduino IDE

• ESP32 Libraries

• Optional Mobile App (Blynk / Web Server)

Semester Wise Plan:

Semester 2 (PC122): Basic automation using PIR + relay + ESP32, proposal submission.

Semester 3 (PC223): Add WiFi control, LDR, LCD display and final demonstration.

Applications:

• Smart classrooms

• Corridors and washrooms

• Homes

• Offices

Learning Outcomes:

Students gain knowledge of embedded programming, IoT basics, sensor interfacing, relay control, automation concepts and teamwork.

Future Scope:

• Occupancy counting

• Energy monitoring

• Mobile dashboard

• Voice control integration

• Full home automation expansion

Conclusion:

The Smart Room Automation System provides a practical solution to energy wastage while giving students real-world experience in IoT and embedded systems.

The project is scalable, cost-effective, and suitable for first-year engineering students.